ASSESSMENT BRIEF COURSEWORK 2

Programme	MSc Data Science		
Module Title	Machine Learning		
Module Code	CMP-L015-0		
Module Level	7	**Assessment Type(s)**	Report and demo
Word Length / Duration		**% contribution to module mark**	40%
Deadline (date & time) for Submission	13 Mar 2026	**Format/Location of submission**	https://moodle.roehampton.ac.uk/mod/assign/view.php?id=2548512
Assessment Feedback date:	27 Mar 2026		

Learning Outcomes

This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the learning outcomes listed below.

LO1: Apply machine learning algorithms to classify images

LO2: Evaluate the performance of a simple learning system using a real-world dataset.

Employability and Professional Skills

This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the following skill(s) which is(are) critical in professional context and jobs.

By successfully completing this assessment, you will be able to

- Apply supervised learning methods to complex classification tasks.
- Implement Neural Network architectures for structured and image data.
- Compare performance of different algorithms and interpret results critically.

LO3: Compare and contrast decision making techniques, such as decision trees, neural networks, and Long Short-Term Memory (LSTM) networks.	

Assessment Requirements

Coursework task

Imagine you work for an AI company developing a machine learning product that classifies images**.** You are asked to design and evaluate a neural network that can learn from an image dataset and accurately predict image categories.

To complete this task, follow the steps below:

1. **Find a Public Image Dataset**

    - Choose a dataset containing labelled images (e.g., animals, flowers, vehicles).
    - You may use any publicly available dataset except for the *digits image dataset* used in the lab solutions.
    - Example resource: https://paperswithcode.com/datasets?task=image-classification

2. **Import and Prepare the Data**

    - Load the dataset into your program.
    - Define input and output variables.
    - Split data into training and testing sets.

3. **Design a Neural Network**

    - Include at least an input layer**,** one or more hidden layers**,** and an output layer**.**
    - You may experiment with layer types, activation functions, optimisers, and architectures (e.g., CNN).

4. **Model Training and Evaluation**

    - Train your neural network on the training dataset.
    - Evaluate model performance on the testing dataset using various metrics.
    - Present your evaluation clearly using tables and/or plots.

5. **Explain the Training Process and Model Structure**

    - Describe your model architecture and training pipeline in clear, simple terms.
    - Use a diagram, flowchart, or model summary plot to illustrate the process.

6. **Optional Enhancements**

- Reproducibility: Rerun your program to demonstrate consistent results.
- Interactive testing: Allow the user to upload or select an image for real-time prediction.
- Additional features: Apply image augmentation, dropout, regularisation, or visualise feature maps.

Submission Requirements

Code Submission:

- Submit your Python file(s) or Jupyter Notebook(s) as StudentID_MLCW2.ipynb.
- Code must be well-commented and logically organised.

Report Submission (PDF maximum one-page A4 size excluding references):

- Your report must move beyond procedural steps. Rather than describing *what* you did in a list, focus on why each methodological choice was made, and provide a critical analysis of the results obtained. Justify your dataset selection, model architecture, hyperparameters, and evaluation metrics with clear reasoning and, where appropriate, evidence from the literature. Discuss strengths, limitations, and sources of error. You are encouraged to consult peer-reviewed research for models of strong critical analysis—e.g., papers from reputable publishers such as Elsevier, Springer, and IEEE—and to situate your work within that scholarly context. All sources must be cited correctly using the IEEE referencing style**.**

Video Submission:

- Record your screen (≤ 10 minutes) while running the code and explaining model design and key outputs.
- Include the video clip (file) in your report.

Demonstration Requirement:

- You should demonstrate your work to your tutor during a scheduled lab session within two weeks after the due date.
- Q & A section for general understanding by answering questions: one or more about the Python code used in your work, and one or more about your understanding of the concepts applied. Please note that you will be marked as *NOT PASS* if you fail to answer the questions.

Marking Criteria (90% for artefact; 10% for report)

Grading

Grade	Criteria

100	Excellent work with extra features can be used in the real-world applications with excellent performance in Q&A	
92	Excellent work with extra features and minor improvement can be used in the real-world application with excellent performance in Q&A	
85	Excellent work with extra features and some improvement can be used in the real-world application with excellent performance in Q&A	
82	Excellent work with extra features and improvement can be used in the real-world application with excellent performance in Q&A	
78	Excellent work with extra features and major improvement can be used in the real-world application with excellent answers in Q&A	
72	Good work completes the required tasks with some additional features that are useful to the real-world applications with good performance in Q&A	
68	Good work completes the required tasks with no bugs and code can be used for similar scenarios with excellent performance in Q&A	
65	Good work completes the required tasks with no bugs with good performance in Q&A	
62	Good work completes the required tasks with minor bugs with good performance in Q&A	
58	Good work completes the required tasks with some bugs with sufficient performance in Q&A	
55	Complete most required tasks with some adequate performance in Q&A	
52	Complete major required tasks with adequate performance in Q&A	
48	Complete some required tasks without bugs with poor performance in Q&A	
45	Complete half required task with some minor bugs in the complete tasks with poor performance in Q&A	
42	Complete half required task with some bugs in the complete tasks with poor performance in Q&A	
38	Does not complete less than half the required tasks with poor performance in Q&A	
35	Does not complete less than half required tasks and some bugs in the completed tasks with poor performance in Q&A	
25	Does not complete less than half required tasks and a lot of bugs in the completed tasks with poor performance in Q&A	
15	Submission with 1 task completed with poor performance in Q&A	

5	Submission without any tasks completed with poor performance in Q&A
0	No submission

Assessment Success Guidance

To achieve maximum success in this assessment, students should go beyond simply meeting the basic requirements. High marks will be awarded for:

- Clarity and depth of analysis – demonstrate strong critical thinking when comparing models, not just reporting results.
- Professional presentation – ensure your code is clean, well-commented, and your report is concise, well-structured, and visually clear.
- Use of additional techniques – applying optional features such as variable importance analysis, cross-validation, or advanced visualisations can help showcase deeper understanding.
- Independent learning – drawing on online resources, tutorials, and academic references to support your approach.

Academic Misconduct:

"Academic integrity and honesty are fundamental to the academic work you produce at the University of Roehampton. You are expected to complete coursework which is your own and which is referenced appropriately. The university has in place measures to detect academic dishonesty in all its forms. If you are found to be cheating or attempting to gain an unfair advantage over other students in any way, this is considered academic misconduct, and you will be penalised accordingly."

Further details about "Student Code of Conduct" and "Disciplinary Regulations" can be found at:

https://www.roehampton.ac.uk/corporate-information/policies/

Use of Artificial Intelligence (AI)

Please read and apply:

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4a%20AI%20principles%20and%20guidance%20-for%20Senate.docx

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4b%20Student%20Guidelines%20on%20the%20use%20of%20AI_.docx

The assessment is designed so that the use of AI during the assessment is possible. You must acknowledge any use of AI and appropriately cite all AI generated outputs. Please

make sure you read and understand the assessment guidelines and ask your Module Leader if you have any questions.

**You can find the student guidance on the use of AI in the Library and Nest*

Referencing

References: Referencing any point in your discussion section(s). If you have used ideas from anywhere other than the lecture notes and tutorial examples (e.g., from a book, the Internet, or a fellow student) then include a reference showing where the code or ideas came from and label your code carefully to show which parts are your and which parts are borrowed.

You should include your reference list at the end of your report. You must ensure you use the appropriate IEEE format for your subject area.

Your referencing must use the IEEE referencing style IEEE Citation Guidelines.

It is highly recommended that you use reference management software such as RefWorks that is provided by the university. Your report should have as many references as is required.

It is your responsibility to ensure that you have actually read all the material you reference, and that the references provided in your report are legitimate and NOT AI-generated.

Mitigating circumstances/late penalties

Sometimes circumstances outside of your control may affect your studies and might prevent you from submitting work on time or attending an exam.

The University offers the ability for students to request additional time to complete an assessment or to defer an examination to a later date. If you are finding yourself in such a situation, please speak to your Academic Guidance Tutor, the Roehampton Student Union (RSU) or someone in the **Wellbeing team** first, who can support you. Further details can be found on the mitigating circumstances portal.

If you do not apply for or are not approved for Mitigating Circumstances, late penalties will apply. If work is submitted up to 14 days late, the mark will be capped at 40%; if it is over 14 days late, it will not be marked.

Resubmissions and Reassessment

If you are required to resubmit this assessment or take part in reassessment, you will be notified via Moodle and your student email. Please ensure you check both regularly. Any reassessment tasks will follow the same learning outcomes and criteria.

Submission Checklist

Before you submit, ask yourself:

Have I fully answered the assessment brief?

Have I met the word count and formatting requirements?

Is my referencing complete and accurate?

Have I declared any AI use honestly?

Have I proofread my work?

Am I submitting through the correct platform before the deadline?